

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Carbon Nanotube Transistors

KKKC6353 Nanoelectronic Devices — Lecture 4, Document 2 of 4

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Table of Contents

The argument of this document **3**
Recap: graphene and nanotubes **4**
1 Four devices that set the agenda, 1998–2006 ..... **5**
1.1 1998: the first nanotube transistors 5
1.2 2001: n-type devices and the first logic 5
1.3 2002–2003: high-κ gates and ohmic contacts 6
1.4 2006: a circuit along a single tube 6
2 Geometry and the two modes of operation **7**
2.1 Gate geometries 7
2.2 Schottky-barrier CNTFETs 8
2.3 MOSFET-like CNTFETs 9
2.4 Tunnel CNTFETs 10
3 Electrostatics: the gate and the quantum capacitance **10**
3.1 The insulator capacitance 10
3.2 The quantum capacitance 11
3.3 Two limits 11
3.4 The natural length of a wrapped tube 12
4 The ballistic transistor: top-of-the-barrier model **13**
4.1 The model 13
4.2 What the model says 14
4.3 Limits of the ballistic picture 14
4.4 The Schottky-barrier model used in Figure 2.6 15
5 Contacts: the resistance budget **15**
5.1 The floor and the overhead 15
5.2 Contact-length scaling 16
6 Doping, CMOS and stability **17**
6.1 Setting the polarity 17
6.2 Hysteresis, sensitivity and passivation 17
7 How short can the gate be? **18**
8 From one tube to integrated circuits **19**
8.1 Where variability comes from 19
8.2 Imperfection-immune design 19
8.3 Wafer-scale arrays and current performance 20
9 Where CNTFETs stand in 2026 ..... **20**
10 Summary **21**
Corrections to the old notes 22
Words to know 22
In-class animations **23**
Sources and further reading **23**

The argument of this document

Lecture 1 ended with a design rule: the channel must be thin, and wrapped by the gate, for the gate to keep control of the barrier. A semiconducting carbon nanotube is about the thinnest channel there can be, a single sheet of atoms 1–2 nm across, with no dangling bonds at its surface and room for a gate all round it. Document 1 showed that it also has a light, symmetric effective mass, a long mean free path and a band gap that can be chosen by choosing the diameter. On paper it is the ideal transistor channel.

Lecture 3's preview table set the CNTFET beside the MOSFET and the single-electron transistor in one line each. This document opens that line up. It makes five points:

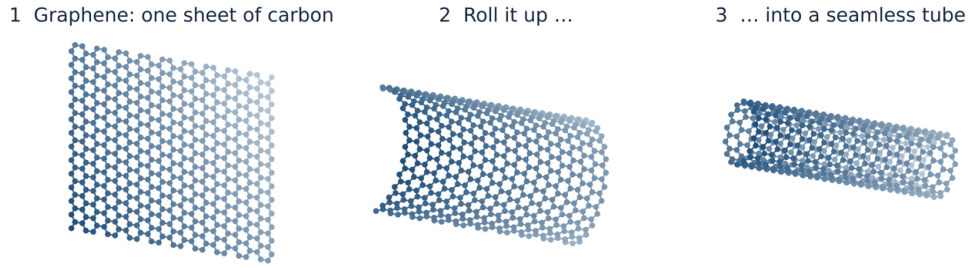
1. A nanotube transistor can work in two different ways. With metal contacts the current is controlled by **tunnelling through Schottky barriers** at the contacts. With doped contacts it is controlled by the **height of a barrier** in the channel, as in a MOSFET. The second is better, and the reasons are physics, not processing.
2. Because the tube is so thin, its own density of states adds a **quantum capacitance** in series with the gate. With a thin high-κ gate the device reaches the quantum-capacitance limit, where the gate moves the bands almost one-for-one.
3. A **ballistic model** with no fitted parameters, the top-of-the-barrier model, predicts the current. It sets the target: about 40 μA per tube at 0.5 V.
4. Real devices fall short mainly because of **contacts**. A single tube has a resistance floor of $h/4e^2 = 6.45$ kΩ, and the metal must transfer electrons into it over a very short length.
5. The physics was settled by about 2012. Since then the problem has been **material and manufacturing**: metallic tubes, pitch and diameter control, doping, and integration. Section 8 traces the progress from a 178-transistor computer (2013) to wafer-scale arrays and a tensor processor (2024), and Section 9 says where things stand in 2026.

Notation follows Document 1: d is the tube diameter, $E_g = 2a_{cc}\gamma_0/d \approx 0.77$ eV nm$/d$, $\hbar v_F = 0.575$ eV nm, and $U_T = kT/q = 25.9$ mV at 300 K. Currents are **per tube** unless stated; arrays are quoted **per μm of width**.

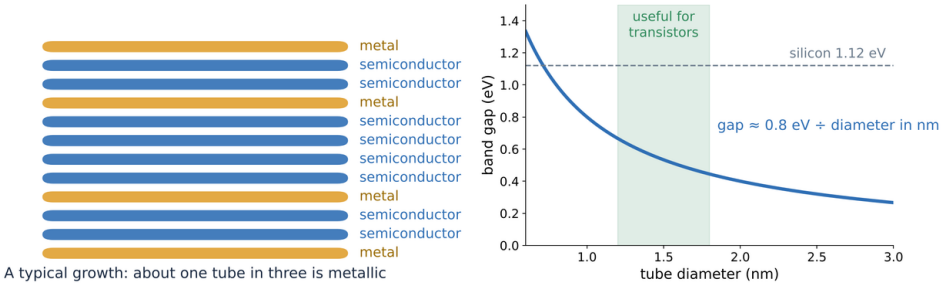
Recap: graphene and nanotubes

> **Four facts from Document 1**
>
> Graphene is a single sheet of carbon atoms in a honeycomb. It conducts superbly, but it has no band gap, so a graphene transistor cannot be switched off.
>
> A carbon nanotube is that sheet rolled into a seamless cylinder 1–2 nm across, with no edges and no dangling bonds.
>
> How the sheet is rolled, given by the chiral indices (n, m), decides the type: when $n - m$ is divisible by 3 the tube is a metal, otherwise a semiconductor. About one tube in three is metallic.
>
> The gap of a semiconducting tube is about 0.8 eV divided by the diameter in nm ($E_g \approx 0.77$ eV nm$/d$): about 0.5 eV for a 1.5 nm tube, half of silicon's.

These four facts are all this document needs from Document 1, which derives them from the band structure of graphene. The first two make the nanotube an almost ideal channel: thin, perfect at its surface, and easy to wrap with a gate. The last two set the practical problems of Sections 2, 7 and 8: the diameter sets the gap and so the leakage, and the metallic tubes that every growth produces cannot be switched off.



Rolling graphene into a nanotube: a flat sheet, part-rolled, and a seamless (14,0) tube. Atom positions computed for this course.



Left: a typical growth mixes metallic and semiconducting tubes, about one in three metallic. Right: the band gap falls as the diameter grows; transistors use tubes of 1.2–1.8 nm. Figures made for this course.

1 Four devices that set the agenda, 1998–2006

> **Key ideas**
>
> The first CNTFETs (1998) were single tubes on a thick oxide with a silicon back gate. They worked, but only as p-type devices, needed volts to switch, and had resistances of megaohms.
>
> By 2006 the field had n-type devices, high-κ top gates, ohmic contacts, logic gates and a ring oscillator built along one tube.
>
> The lessons of those years shaped every device since: the contacts dominate, and the environment of the tube matters.

1.1 1998: the first nanotube transistors

Two groups reported nanotube transistors in 1998: Tans, Verschueren and Dekker at Delft and Martel and co-workers at IBM. Both laid single-walled tubes from solution across metal electrodes on an oxidised, heavily doped silicon wafer, which served as the gate (Figure 2.1). The conductance could be changed by more than five orders of magnitude, but only by applying several volts across 100–300 nm of SiO_2. The devices conducted when the gate was *negative*, that is, they were **p-type**. On-resistances were in the megaohms, and the current did not saturate.

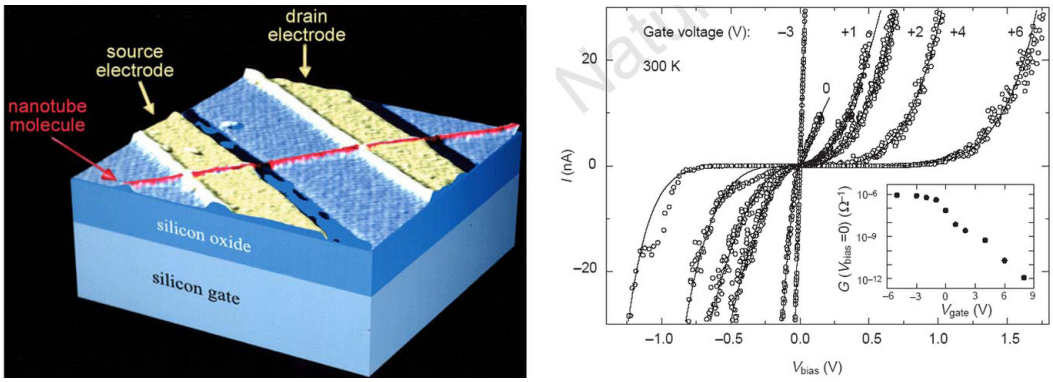


Figure 2.1. *The first carbon nanotube transistor. Left: a single nanotube lies across two platinum electrodes on SiO_2, with the silicon substrate as the gate. Right: current against bias at several gate voltages; the inset shows the conductance falling by more than five orders of magnitude as the gate voltage rises from −6 to +9 V. Source: S. J. Tans, A. R. M. Verschueren and C. Dekker, Nature 393, 49 (1998); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).*

Why p-type? The tubes were undoped. It turned out that the polarity was set mainly at the **contacts**: oxygen adsorbed from the air on the metal–tube contacts changes the local work function and the Schottky barrier, favouring hole injection (Heinze *et al.*, 2002). This was the first sign of a theme that runs through the subject: in a CNTFET the contacts are not passive wires, they *are* much of the device.

1.2 2001: n-type devices and the first logic

Derycke and co-workers at IBM (2001) made n-type devices in two ways: by annealing in vacuum to remove the adsorbed oxygen, and by doping the tube with potassium. Covering one half of a tube with resist and annealing the other gave a p-type and an n-type transistor on the **same tube**, which they wired as a complementary inverter with voltage gain (Figure

2.2). In the same year Bachtold, Hadley, Nakanishi and Dekker at Delft used local aluminium gates under the tubes, with a thin Al_2O_3 insulator, and built an inverter, a NOR gate, a static memory cell and a ring oscillator from one to three CNTFETs each.

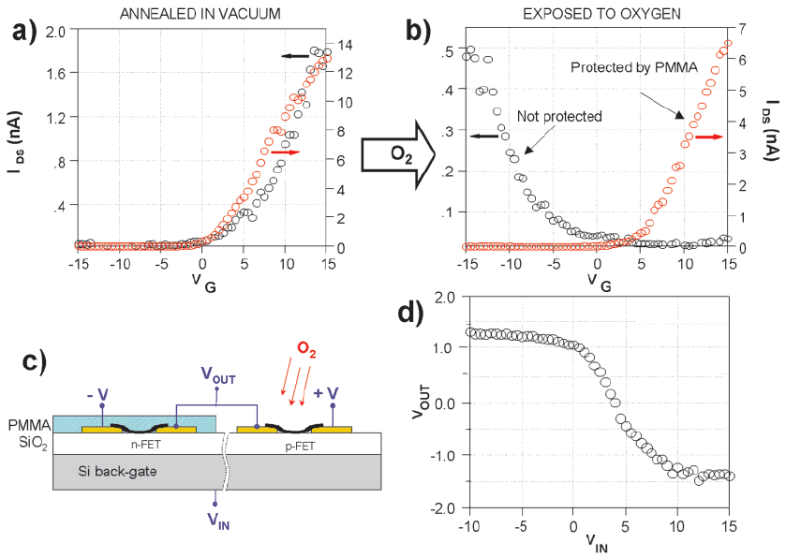


Figure 2.2. *The first CMOS-like nanotube inverter. (a) After annealing in vacuum, both devices are n-type. (b) Exposing one to oxygen while the other is protected by PMMA turns it p-type. (c, d) The two wired as an inverter, with gain above 1. Source: V. Derycke, R. Martel, J. Appenzeller and Ph. Avouris, Nano Letters 1, 453 (2001); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).*

1.3 2002–2003: high-κ gates and ohmic contacts

Moving the gate on top of the tube, on a few nanometres of insulator, brought the switching voltage down to about 1 V (Wind *et al.*, 2002). Javey and co-workers (2002) used 8 nm of ZrO_2 ($\kappa \approx 25$), reaching a subthreshold swing of about 70 mV/dec. In 2003 the same group found that **palladium** forms almost barrier-free contacts to the valence band of tubes wider than about 1.5 nm. The resulting p-type devices were nearly ballistic, with on-state conductances approaching the $4e^2/h$ limit (Javey *et al.*, 2003). In 2007 **scandium** (and later yttrium) contacts did the same for the conduction band (Zhang *et al.*, 2007). This gave n-type and p-type devices of almost equal strength, something silicon has never had.

1.4 2006: a circuit along a single tube

Chen and co-workers at IBM (2006) built a five-stage ring oscillator along one 18 µm nanotube, using palladium and aluminium gates to make the p-type and n-type devices (Figure 2.3). It oscillated at 52 MHz, limited by the parasitic capacitance of the wiring rather than by the tube. It showed that nanotube circuits could be fast, and that wiring and integration, not the transistor, would be the hard part.

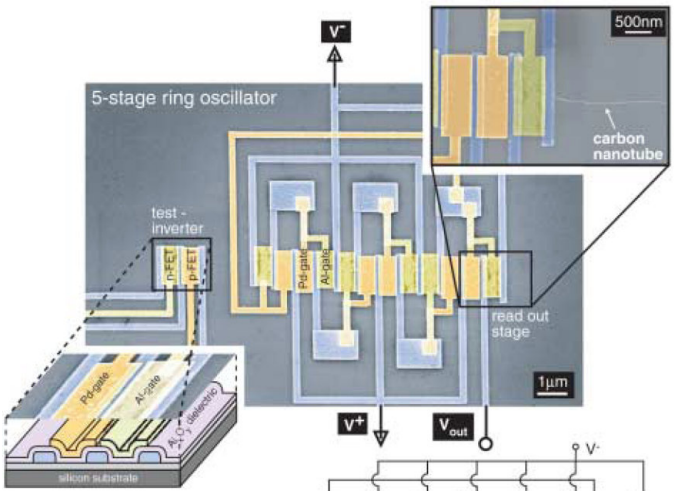


Figure 2.3. *A five-stage ring oscillator built along a single nanotube. The tube runs horizontally under alternating Pd (p-type) and Al (n-type) gates. Source: Z. Chen et al., Science 311, 1735 (2006); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).*

Check your understanding

1. Why were the first CNTFETs p-type, although the tubes were undoped?
2. What did palladium and scandium contacts make possible that silicon cannot easily do?

2 Geometry and the two modes of operation

> **Key ideas**
>
> CNTFETs are built with back, top or wrap-around gates, on one tube or on aligned arrays of many.
>
> With metal source and drain the gate controls the *thickness* of Schottky barriers at the contacts: a tunnelling transistor. With doped source and drain it controls the *height* of a barrier in the channel: a MOSFET-like transistor.
>
> Schottky-barrier devices conduct both electrons and holes (ambipolar) and switch less steeply. MOSFET-like devices behave like ideal MOSFETs.

2.1 Gate geometries

Figure 2.4 summarises the geometries. A nanotube is not bound to a substrate, so any gate geometry can be built around it. The back gate of 1998 is easy to make but couples weakly and switches every device on the wafer at once. Local top gates on high-κ insulators give each device its own gate and a thin insulator. The ideal is the **gate-all-around** (coaxial) geometry, which Lecture 1 showed has the shortest natural length. For real circuits one tube is not enough current, so practical devices use **aligned arrays**: many parallel tubes, ideally 100–200 per µm, under a common gate.

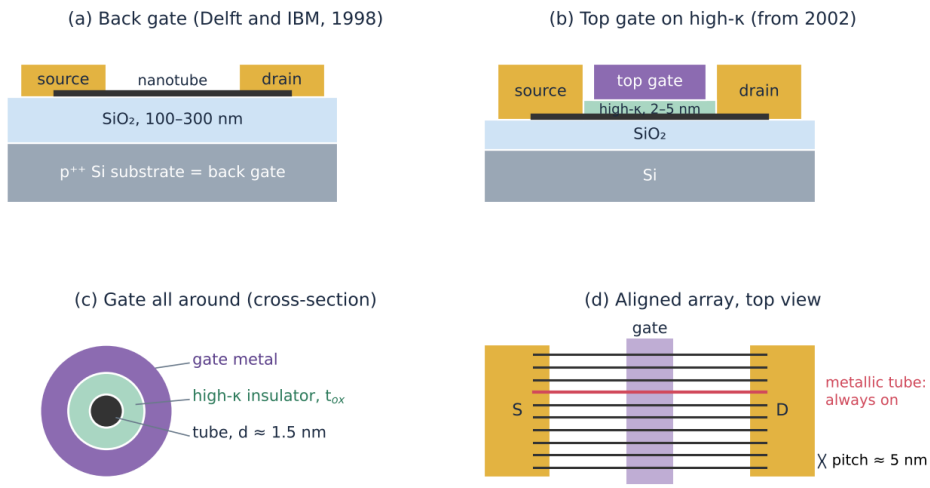


Figure 2.4. *CNTFET geometries. (a) The 1998 back-gated device. (b) A local top gate on a thin high-κ insulator. (c) Gate all around, the electrostatic ideal. (d) An aligned array for a practical device: many tubes in parallel under one gate, where a single metallic tube (red) cannot be switched off. Figure made for this course; not to scale.*

2.2 Schottky-barrier CNTFETs

When the source and drain are metal, there is a **Schottky barrier** at each contact. Its height for electrons is set by the alignment of the metal Fermi level with the tube's conduction band. In nanotubes the barrier height follows the metal's work function more closely than in silicon, because a one-dimensional contact has little area for interface states to pin the Fermi level (Léonard and Tersoff, 2000).

The gate does not change the barrier *height*; it changes its *width*. The bands in the channel follow the gate, but at the contact they are pinned to the metal, so the band bends over a distance set by the electrostatics, roughly the natural length λ (Lecture 1), which for a thin wrap-around gate is a few nanometres. As the gate voltage rises the barrier thins, and electrons **tunnel** through it (Figure 2.5, top). The device is a tunnelling transistor, and Lecture 3's WKB physics sets its current.

Three consequences follow, all visible in Figure 2.6:

- **Ambipolar conduction.** If the metal Fermi level sits near midgap, a low gate voltage thins the barrier for *holes* at the drain end. The device then conducts both electrons and holes, giving a V-shaped transfer curve. The minimum current, where both barriers are thick, is the off-state.
- **Diameter sets the off-current.** The minimum current falls roughly as $\exp\left(-E_g/2U_T\right)$ for a midgap barrier, so wider tubes, with smaller gaps, leak much more. In Figure 2.6(a) going from 1.0 to 2.0 nm raises the minimum current by three orders of magnitude.
- **The insulator sets the swing.** Thinner, higher-κ insulators shorten λ and thin the barrier faster with gate voltage (Figure 2.6(b)). With thick oxides the subthreshold swing of an SB-CNTFET is far worse than 60 mV/dec (Guo, Datta and Lundstrom, 2004).

> **A correction to the old notes.** The Gwalior slides conclude that "increasing CNT diameter has good effect on CNTFET". Wider tubes do give more on-current (smaller

> barriers, lower contact resistance and higher mobility), but they also give much more off-current and ambipolar leakage. Their simulated table shows the leakage *unchanged* from 0.5 to 5 nm, an artefact of the simulation set-up rather than physics. The useful window is about 1.2–1.8 nm.

2.3 MOSFET-like CNTFETs

If the tube segments between the gate and the contacts are heavily **doped** (chemically, electrostatically by a second gate, or by a charged dielectric), the contacts become ohmic, and the gate instead controls the **height** of a barrier in the middle of the channel (Figure 2.5, bottom). This is exactly how a MOSFET works: thermionic emission over a gate-controlled barrier. The consequences reverse:

- The device is **unipolar**: n-type with n^+ extensions, p-type with p^+.
- The subthreshold swing can approach the thermal limit $\ln(10)\, U_T = 60$ mV/dec at room temperature.
- The off-current is limited by thermionic emission over the barrier and, at high drain bias, by **band-to-band tunnelling** between the channel and the drain, which again worsens as the gap shrinks.

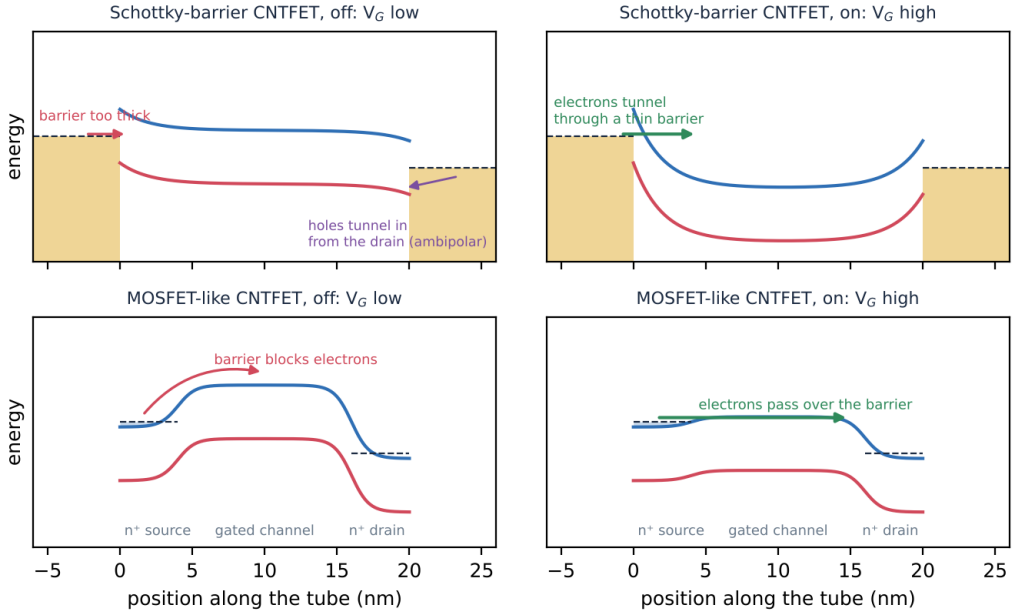


Figure 2.5. *Band diagrams along the tube. Top: a Schottky-barrier CNTFET; the gate thins the barriers at the metal contacts (gold), and electrons tunnel through. At low gate voltage holes can tunnel in from the drain. Bottom: a MOSFET-like CNTFET with doped extensions; the gate lowers a barrier in the channel, and electrons go over it. Blue: conduction band; red: valence band; dashed: Fermi levels. Figure made for this course; computed profiles, schematic energies.*

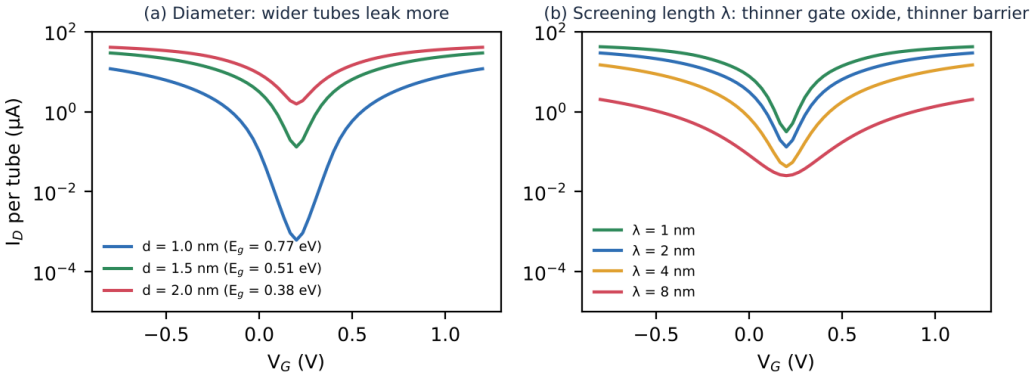


Figure 2.6. *Schottky-barrier CNTFET transfer curves, from a Landauer calculation with WKB tunnelling through the tube's band gap, a midgap barrier, a 20 nm channel and V_D = 0.4 V. (a) Wider tubes leak far more at the minimum. (b) A shorter screening length λ (thinner insulator) gives steeper switching. Figure made for this course; the model is described in Section 4.4.*

2.4 Tunnel CNTFETs

A third mode uses the tunnelling of Lecture 3 deliberately. With a p^+ source and an n-type channel, the gate pulls the channel's conduction band below the source's valence band, and electrons tunnel across the gap: a **tunnel FET**. Appenzeller and co-workers (2004) measured a swing of 40 mV/dec in a nanotube tunnel FET, below the 60 mV/dec limit, but at very low current, which is the usual difficulty with tunnel FETs (Lecture 3, Document 2).

Check your understanding

1. In an SB-CNTFET, what does the gate voltage change: the height or the width of the contact barrier?
2. Why is an SB-CNTFET with midgap contacts ambipolar, and how does the minimum current depend on diameter?
3. Give two ways of making the contacts of a CNTFET ohmic.

3 Electrostatics: the gate and the quantum capacitance

> **Key ideas**
>
> A wrap-around gate has a capacitance per length $C_{ins} = 2\pi\varepsilon/\ln[(2t_{ox} + d)/d]$.
>
> The tube's own density of states adds a quantum capacitance $C_Q \approx 0.36$ fF/μm in series.
>
> With a thin high-κ gate, $C_{ins} > C_Q$: the gate moves the tube's bands almost one-for-one (ideal switching), and the charge is set by the tube's density of states (the quantum-capacitance limit).

3.1 The insulator capacitance

For a tube of diameter d inside a coaxial gate with insulator thickness t_{ox} and permittivity $\varepsilon = \kappa\varepsilon_0$, the capacitance per unit length is that of a coaxial cable:

$$C_{ins} = \frac{2\pi\varepsilon}{\ln\left(\frac{2t_{ox}+d}{d}\right)} = \frac{2\pi\varepsilon}{\ln\left(\frac{t_{ox}+r}{r}\right)}$$

where $r = d/2$ is the tube *radius*. (The Gwalior slides write this with r_{cnt} and call it the diameter; it is the radius.) A top gate on a planar insulator gives somewhat less, roughly $2\pi\varepsilon/\ln(4t_{ox}/d)$ for $t_{ox} \gg d$ with the image in the gate.

3.2 The quantum capacitance

Adding charge to the tube does two things. It changes the electrostatic potential, as in any capacitor, and it fills states in the tube's bands, which costs energy because the density of states is finite. The second effect is a capacitance in series:

$$C_Q = e^2\, D(E_F), \qquad C_G = \left(\frac{1}{C_{ins}} + \frac{1}{C_Q}\right)^{-1}$$

In a metallic tube, or a semiconducting tube with the Fermi level well into its first subband, $D = 8/hv_F$ (Document 1, Section 5.2), so

$$C_Q = \frac{8e^2}{hv_F} \approx 3.6 \times 10^{-10}\ \text{F/m} = 0.36\ \text{fF/μm}$$

(often rounded to 0.4 fF/μm, the value for $v_F = 8 \times 10^5$ m/s). Near a subband edge the van Hove peak makes C_Q larger; in the off-state, with the Fermi level in the gap, $C_Q \to 0$.

3.3 Two limits

The gate voltage divides between the two capacitors. The change in the tube's potential is

$$\Delta\psi_s = \frac{C_{ins}}{C_{ins} + C_Q}\,\Delta V_G$$

- **Classical (MOS) limit, $C_{ins} \ll C_Q$.** Thick or low-κ insulators. The tube's potential moves only a fraction of the gate voltage, and the charge is $C_{ins}(V_G - V_T)$, as in a silicon MOSFET.
- **Quantum-capacitance limit, $C_{ins} \gg C_Q$.** Thin high-κ insulators. The tube's bands follow the gate almost one-for-one, so the subthreshold swing is ideal, and the charge is set by C_Q, that is, by the density of states. Making the insulator thinner still adds almost nothing.

A silicon MOSFET has a quantum capacitance too, but its two-dimensional density of states is so large that it is usually a correction. In a nanotube it is comparable with the insulator capacitance, as Figure 2.7 shows.

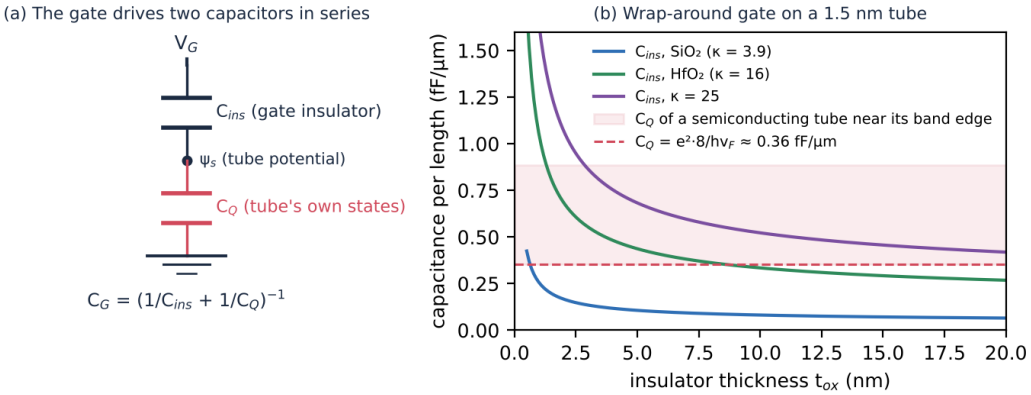


Figure 2.7. *(a) The gate drives the tube through the insulator capacitance and the quantum capacitance in series. (b) Wrap-around insulator capacitance on a 1.5 nm tube against thickness, for three insulators, compared with C_Q. A few nanometres of HfO_2 already exceed C_Q. Figure made for this course from the formulas above.*

> **Worked example: how close to the limit?**
>
> A 1.5 nm tube with 3 nm of HfO_2 ($\kappa = 16$) all round: $C_{ins} = 2\pi \times 16 \times 8.85 \times 10^{-12} / \ln(7.5/1.5) = 5.5 \times 10^{-10}$ F/m $= 0.55$ fF/µm. With $C_Q = 0.36$ fF/µm, $C_G = (1/0.55 + 1/0.36)^{-1} = 0.22$ fF/µm, and the tube's potential follows $0.55/0.91 = 60\%$ of the gate voltage *in the on-state*. In the off-state ($C_Q \to 0$) it follows almost 100%, which is what sets the swing. With 3 nm of SiO_2, $C_{ins} = 0.13$ fF/µm and the device is near the classical limit.
>
> For comparison, a silicon nanosheet with 1 nm EOT has $C_{ox} = 3.45 \times 10^{-6}$ F/cm^2 = 34.5 fF per µm^2 of gate area. An array of 200 tubes per µm, each with $C_G = 0.22$ fF/µm, gives 44 fF per µm^2, about the same. The nanotube array matches silicon's gate charge only if it is dense.

3.4 The natural length of a wrapped tube

Lecture 1 used the natural length λ to judge how short a channel can be. For a cylindrical channel of diameter d inside a coaxial insulator of thickness t_{ox}, Auth and Plummer (1997) derived

$$\lambda_{GAA} \approx \sqrt{\frac{2\varepsilon_{ch} d^2 \ln(1 + 2t_{ox}/d) + \varepsilon_{ox} d^2}{16\,\varepsilon_{ox}}}$$

For a 1.5 nm tube with 3 nm of HfO_2, taking the tube's permittivity to be similar to the insulator's, this gives $\lambda \approx 0.8$ nm. Even with generous allowances for the uncertainty in ε_{ch}, λ is about 1 nm, against 2–3 nm for a silicon nanosheet. By the $L \gtrsim 5\lambda$ rule, gate lengths down to about 5 nm should be free of short-channel effects. Experiment agrees (Section 7). Note that λ here describes the channel; the Schottky-barrier width of Section 2.2 is set by the same electrostatics but also by the geometry of the contact, and is usually a little longer.

Check your understanding

1. Compute C_{ins} for a 1.2 nm tube with 5 nm of SiO_2 all round. Is the device near the classical or the quantum-capacitance limit?

2. Why does C_Q fall to zero in the off-state, and why does that help the subthreshold swing?

4 The ballistic transistor: top-of-the-barrier model

> **Key ideas**
> In a short tube most electrons cross without scattering. The current is then set by how many states at the top of the barrier are filled from the source and how many from the drain.
> Two equations, one for charge and one for current, solved together, give the whole I–V characteristic with no fitted mobility.
> The result: about 60 mV/dec, saturation at low drain voltage, and roughly 40 μA per tube at 0.5 V.

4.1 The model

The model is due to Natori (1994) for silicon and was developed for nanotubes by Rahman, Guo, Datta and Lundstrom (2003). Focus on the top of the barrier in the channel. States there moving towards the drain ($+k$) are filled from the source, with Fermi level μ_S; states moving towards the source ($-k$) are filled from the drain, with $\mu_D = \mu_S - qV_D$. The electron density is

$$n = \int D(E)\,\frac{f(E+U-\mu_S)+f(E+U-\mu_D)}{2}\,\mathrm{d}E$$

where U is the shift of the band at the top of the barrier. U is set by the gate and drain through capacitive coupling, and by the charge itself:

$$U = -q(\alpha_G V_G + \alpha_D V_D) + \frac{q^2 n}{C_\Sigma}$$

with $\alpha_G = C_{ins}/C_\Sigma$, $\alpha_D = C_D/C_\Sigma$, and $C_\Sigma = C_{ins} + C_S + C_D$ per unit length. The two equations are solved together for U. The current is the difference between the right-going and left-going fluxes. For a one-dimensional subband with four channels every electron that enters is carried at its group velocity, and the density of states and velocity cancel:

$$\boxed{I = \frac{4qU_T}{h}\,q\left[\ln\left(1+e^{(\mu_S-E_1-U)/kT}\right)-\ln\left(1+e^{(\mu_D-E_1-U)/kT}\right)\right]}$$

summed over subbands with edges $E_1, E_2, \ldots$ Two limits are worth remembering:

- **Subthreshold** (barrier well above μ_S): $I \propto e^{-U/kT}$, and if $\alpha_G \approx 1$ the swing is $\ln(10)\,U_T = 60$ mV/dec.
- **On-state, V_D large**: the drain term vanishes, and with the barrier pulled well below μ_S, $I \to (4q^2/h)(\mu_S - E_1 - U)/q$. A single subband can carry at most $4q^2/h = 155$ μA per volt of overdrive.

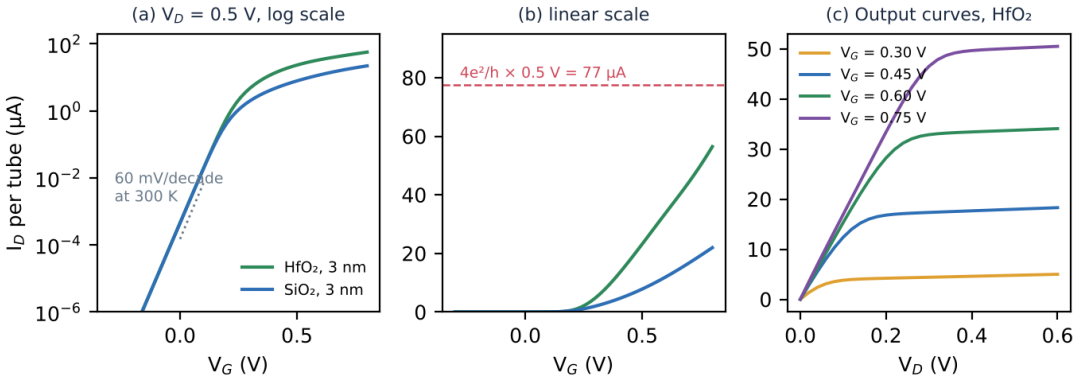


Figure 2.8. *Top-of-the-barrier ballistic model for a 1.5 nm tube with a wrap-around gate and doped contacts, two subbands. (a) Transfer curves at V_D = 0.5 V on a log scale: 60 mV/dec below threshold for both insulators. (b) The same on a linear scale: HfO_2 gives about three times the on-current of SiO_2. (c) Output curves with HfO_2: the current saturates once V_D exceeds a few kT/q. Figure made for this course; model after Rahman et al. (2003).*

4.2 What the model says

Figure 2.8 shows the model for a 1.5 nm tube. Four results matter:

- **Ideal subthreshold swing.** With a thin high-κ wrap-around gate, $\alpha_G \approx 0.97$ and the swing is 60 mV/dec.
- **Early saturation.** The current saturates once V_D is a few kT/q above zero, as soon as the drain no longer fills the left-going states. There is no pinch-off; the saturation is a property of ballistic injection.
- **About 40 μA per tube** at 0.5 V of overdrive. The quantum capacitance limits how much charge the gate can add, so the current per tube approaches, but does not reach, $4q^2/h \times$ overdrive.
- **Injection velocity.** The average velocity of the injected electrons is 2–4 × 10^5 m/s, two to four times that of a silicon nanosheet. This, not mobility, is the figure of merit of a ballistic channel.

> **Worked example: from tubes to mA/μm**
>
> At 40 μA per tube and a pitch of 5 nm (200 tubes per μm), an ideal array would carry 8 mA/μm. A 2026 silicon nanosheet transistor delivers about 1–2 mA/μm at 0.7 V. The best measured CNT arrays reach 1–3 mA/μm (Section 8). The difference between 8 and 2 mA/μm goes to contact resistance, tubes that are not perfectly aligned or evenly spaced, screening between neighbouring tubes (which lowers the gate capacitance per tube in a dense array), and interface traps. These, not the tube itself, are what engineering must now fix.

4.3 Limits of the ballistic picture

Ballistic transport is a good approximation because channels of 10–100 nm are much shorter than the low-bias mean free path of about 1 μm. But at high drain bias, electrons near the drain gain enough energy to emit optical phonons (Document 1, Section 6.2), which return some of them to the source. Real devices reach 50–80% of the ballistic limit. The full treatment uses the non-equilibrium Green's function method coupled to Poisson's

equation, as in the diameter study in the old notes; the top-of-the-barrier model captures its main results with a fraction of the effort.

4.4 The Schottky-barrier model used in Figure 2.6

For the Schottky-barrier device the barrier is at the contact, so the top-of-the-barrier model does not apply. Figure 2.6 instead uses a Landauer current

$$I = \frac{4q}{h}\int \mathcal{T}(E)[f_S(E) - f_D(E)]\mathrm{d}E$$

with the transmission from WKB (Lecture 3, Document 1): $\mathcal{T} = \exp\left(-2\int \kappa\, \mathrm{d}x\right)$. Inside the gap the tube's two-band dispersion gives an imaginary wavevector

$$\kappa(E,x) = \frac{\sqrt{\Delta^2 - \left(E - E_{mid}(x)\right)^2}}{\hbar v_F}, \qquad \Delta = E_g/2$$

which is largest at midgap and falls to zero at both band edges. The same expression therefore describes tunnelling of electrons near the conduction band and of holes near the valence band, which is why the model gives ambipolar curves. The band profile bends from the contact to the channel over a length λ, $E_{mid}(x) = E_{ch} + (E_{contact} - E_{ch})\, e^{-x/\lambda}$.

Check your understanding

1. Why does the current of a ballistic transistor saturate without pinch-off?
2. Estimate the maximum current one subband can carry at 0.3 V of overdrive.
3. List three reasons a dense array delivers less than the number of tubes times the single-tube current.

5 Contacts: the resistance budget

> **Key ideas**
>
> Each tube has a resistance floor of $h/4e^2 = 6.45$ kΩ, however good its contacts.
>
> Metal laid on top of a tube (a side contact) transfers electrons over a length of 20–50 nm; shorter contacts cost resistance fast. That is a problem when a whole transistor must fit in about 40 nm.
>
> End-bonded contacts, where the metal bonds to the end of the tube, have a resistance independent of contact length.

5.1 The floor and the overhead

The total resistance of a ballistic tube is

$$R_{tube} = \frac{h}{4e^2} + 2R_{c,extra} + R_{channel}$$

The first term is the unavoidable quantum resistance, split between the two contacts. $R_{c,extra}$ is whatever a real contact adds: a residual Schottky barrier, a tunnel gap between metal and tube, or poor wetting. $R_{channel}$ is small for short channels. With palladium on a 1.5 nm tube and contacts longer than about 100 nm, $R_{c,extra}$ can be a few kΩ, so the device is close to the 6.45 kΩ floor.

5.2 Contact-length scaling

In a side contact the metal lies along the tube. Electrons transfer from metal to tube along the overlap, over a **transfer length** L_T of a few tens of nanometres. As with a contact to a silicon resistor, shortening the contact below L_T raises the resistance roughly as $\coth(L_c/L_T)$. Franklin and Chen (2010) measured this: palladium contacts to single tubes degraded sharply below about 20 nm. At an advanced node the whole contacted gate pitch is about 40–50 nm, so the contacts can be 10–20 nm long at most.

IBM's answer was the **end-bonded contact** (Cao *et al.*, 2015). Molybdenum is deposited over the tube end and annealed, so it forms molybdenum carbide and bonds to the open end of the tube. The resistance, about 36 kΩ per tube in total, did not increase as the contact was shortened to 10 nm. Two years later the same group fitted a complete p-type CNTFET (contacts, channel and spacers) into a 40 nm footprint and showed it outperforming silicon devices of the same footprint (Cao *et al.*, 2017).

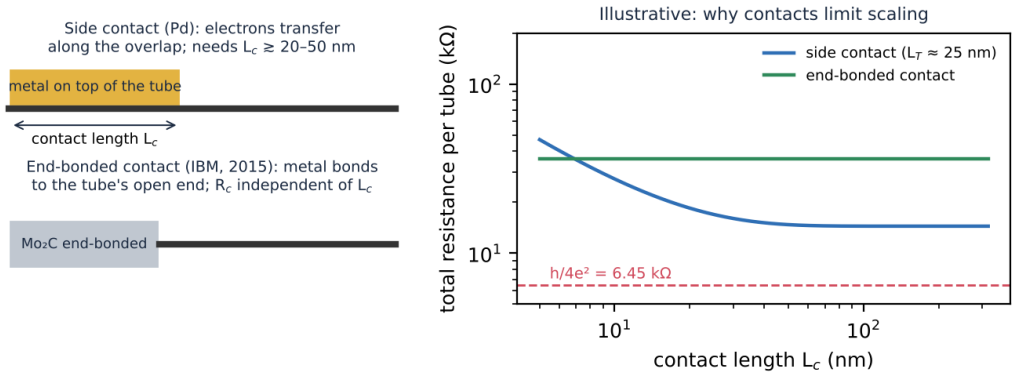


Figure 2.9. *Left: a side contact, where electrons transfer from the metal into the tube along the overlap, and an end-bonded contact, where the metal bonds to the tube's end. Right: total resistance of one tube against contact length (illustrative): the side contact rises steeply below its transfer length; the end-bonded contact is flat but its value is still several times the 6.45 kΩ floor. Figure made for this course after Franklin and Chen (2010) and Cao et al. (2015).*

> **Worked example: resistance per µm**
>
> An array of 200 tubes per µm, each with a total resistance of 20 kΩ, has a resistance of $20\ \text{k}\Omega/200 = 100\ \Omega\cdot\mu\text{m}$. That is comparable with the total series resistance budget of an advanced silicon transistor. The quantum floor alone, $6.45\ \text{k}\Omega/200 = 32\ \Omega\cdot\mu\text{m}$, is not the problem; the extra contact resistance and the number of tubes per µm are.

Check your understanding

1. Why can't better metallurgy remove the 6.45 kΩ term?
2. Why does contact length matter more at advanced nodes than it did in 2003?

6 Doping, CMOS and stability

> **Key ideas**
>
> Substitutional doping, the way silicon is doped, does not work well in nanotubes. Polarity is set instead by the contact metal, by electrostatic doping, or by charge in an adjacent layer.
>
> High-work-function metals (Pd) make p-type devices; low-work-function metals (Sc, Y) make n-type.
>
> Exposed tubes are very sensitive to their surroundings: adsorbed water and oxygen cause hysteresis and threshold drift, so channels must be encapsulated.

6.1 Setting the polarity

Replacing carbon atoms with boron or nitrogen damages the tube and scatters carriers, so CNT electronics sets polarity in other ways:

- **Contact work function.** Pd (work function about 5.1 eV) aligns with the valence band of tubes wider than about 1.5 nm: p-type. Sc (about 3.3 eV) or Y aligns with the conduction band: n-type. This is the cleanest method and the one used for most high-performance devices.
- **Electrostatic doping.** A second gate, or a back gate under the extensions, holds the access regions in accumulation, making them n^+ or p^+ without dopant atoms.
- **Charge-transfer doping.** Molecules adsorbed on the tube (potassium, polyethylenimine, and others) donate or accept electrons. Effective, but often not stable in air.
- **Dielectric doping.** Fixed charge in a deposited layer, such as SiN_x or a non-stoichiometric oxide, shifts the bands of the tube below it. This is how A version of this, combined with contact engineering, was used to make the n-type devices of the 2019 RV16X-NANO processor (the MIXED process; Section 8).

6.2 Hysteresis, sensitivity and passivation

A tube is all surface. Water molecules on the SiO_2 next to it, and charge traps in the insulator, respond slowly to the gate field. The transfer curve then depends on the sweep direction (**hysteresis**), and the threshold drifts (Kim *et al.*, 2003). The same sensitivity makes bare CNTFETs excellent **gas sensors**: NO_2 and NH_3 shift the threshold through charge transfer at the tube body and at the contacts. For logic, the channel is buried in a dense, low-trap dielectric, often deposited by atomic-layer deposition, and in some processes treated with a hydrophobic monolayer to keep water away. The old notes report that such a treatment narrows the spread of threshold voltages by more than half.

> **A correction to the old notes.** The Gwalior slides state that choosing a dielectric for CNTFETs "is not related to Fermi level pinning or passivating surface states". It is true that a nanotube has no dangling bonds, so there is no silicon-style interface to passivate. But traps in the dielectric next to the tube matter a great deal: they cause hysteresis, threshold variation and a worse subthreshold swing. TSMC's 2024 devices (Section 8) name interface traps as the main remaining problem.

Check your understanding

1. Why is substitutional doping rarely used in nanotubes?
2. What causes hysteresis in CNTFETs, and why does it make them good sensors?

7 How short can the gate be?

> **Key ideas**
> CNTFETs have kept good electrostatics down to gate lengths of 9 nm (2012) and 5 nm (2017).
> The best single-tube devices out-perform silicon devices at the same gate length, normalised by tube diameter.
> At these sizes, contacts and the footprint of the whole device, not the channel, decide performance.

Year	Group	Device	Result
2005	Seidel *et al.* (Infineon)	Sub-20 nm channels	Short-channel CNTFETs without short-channel effects
2012	Franklin *et al.* (IBM)	9 nm channel, single tube, high-κ	SS 94 mV/dec; 2.41 mA/μm normalised by tube diameter at 0.5 V
2015	Cao *et al.* (IBM)	End-bonded Mo contacts	Contact resistance independent of contact length down to 10 nm
2017	Qiu *et al.* (Peking)	5 nm gate, graphene contacts	SS about 73 mV/dec; switching faster than silicon devices of the same length
2017	Cao *et al.* (IBM)	Whole device in a 40 nm footprint	Higher current than silicon devices of the same footprint

The pattern is clear. Electrostatics stopped being the problem at about 10 nm, as the natural length predicts (Section 3.4). The Franklin result is quoted per μm of *tube diameter* because a single tube has no width. Turning that number into a real per-μm figure requires packing tubes at a pitch close to their diameter, which is the subject of Section 8.

> **Reading benchmark numbers.** CNT papers quote current per tube, per μm of tube diameter, per μm of array width, or per μm of device width including the gaps between arrays. These can differ by a factor of ten for the same device. Always check which one is meant before comparing with silicon, whose figures are per μm of transistor width.

Check your understanding

1. Why does the natural length predict that CNTFETs can be shorter than silicon FinFETs?
2. A single tube of 1.2 nm carries 25 μA. What is that "per μm of diameter"? What would an array at 5 nm pitch carry per μm of width?

8 From one tube to integrated circuits

> **Key ideas**
>
> Circuits need billions of tubes with the right type, diameter, position and pitch. Every one of those is a source of variability.
>
> Stanford and MIT made complete computers anyway, by designing circuits that tolerate imperfect tubes: 178 transistors in 2013, a 16-bit RISC-V processor with 14,702 CNTFETs in 2019.
>
> Since 2020, CNTFETs have been made in commercial silicon fabs, aligned arrays have reached 100–200 tubes per µm, and both TSMC and Peking University have shown devices and circuits at 2024 performance levels.

8.1 Where variability comes from

Source	Effect	Remedy
Metallic tubes	Cannot be switched off; short the device; ruin I_{on}/I_{off}	Sort to >99.99–99.9999%; electrical burn-off; tolerant design
Diameter spread	E_g, V_T and on-current vary	Narrow-diameter sorting
Pitch and count	Number of tubes per transistor varies	Self-aligned assembly; more tubes per device
Misalignment, bundling	Reduced current; screening; shorts	Dense, well-aligned arrays
Dielectric traps	Hysteresis, V_T shift, worse SS	Low-trap gate stacks, encapsulation
Residues, particles	Opens and shorts in large circuits	Wafer cleaning (RINSE)

8.2 Imperfection-immune design

Shulaker and co-workers at Stanford (2013) built the first nanotube computer: 178 CNTFETs, each with 10–200 tubes, running at 1 kHz. It executed a single instruction (subtract and branch if negative), which is enough to run a simple operating system, and multitasked between counting and sorting. Two techniques made it work. Metallic tubes were removed by **electrical breakdown**: with the semiconducting tubes switched off by the gate, a large current burns out the metallic ones (Collins, Arnold and Avouris, 2001). And the circuit layout was designed so that misaligned or stray tubes could not form wrong connections.

Hills and co-workers at MIT (2019) scaled this up to **RV16X-NANO**, a 16-bit RISC-V microprocessor of **14,702 CNTFETs**, which ran a “Hello, world!” program. Three process and design methods made it possible:

- **DREAM** (designing resiliency against metallic CNTs): logic gates were chosen and sized so that the noise margins survive the metallic tubes in material of 99.99% purity, which is achievable, rather than 99.999999%, which is not.
- **RINSE** (removal of incubated nanotubes through selective exfoliation): a coating and sonication step that strips the bundles and particles which otherwise cause defects in large circuits.

- **MIXED** (metal interface engineering crossed with electrostatic doping): n-type and p-type devices made with different contact metals and a charge-carrying oxide.

The same group built CNTFET logic on top of silicon CMOS and resistive memory on one chip (Shulaker *et al.*, 2017). Nanotubes can be processed below 400 °C, so they can be built *above* finished silicon circuits: **monolithic 3D integration**. In 2020 Bishop and co-workers transferred the process to two commercial silicon fabs (Analog Devices and SkyWater Technology), using standard 200 mm tools.

8.3 Wafer-scale arrays and current performance

Year	Group	Result
2020	Liu *et al.* (Peking)	Aligned arrays on 10 cm wafers, 100–200 tubes per µm, >99.9999% semiconducting; transistors above 1 mA/µm; ring oscillators at 8 GHz
2020	Bishop *et al.* (MIT, Analog Devices, SkyWater)	CNTFET fabrication in commercial silicon fabs
2024	TSMC Research and collaborators	Aligned-CNT channel in a gate-all-around *nanosheet* structure: above 1 mA/µm at V_{DS} = −0.5 V, 70 nm channel, SS 135 mV/dec
2024	Si *et al.* (Peking)	Tensor processing unit of about 3,000 CNTFETs; 2-bit multiply–accumulate; 88% on handwritten digits (MNIST) at 295 µW
2025	Zhou, Pan *et al.* (Peking, Xiangtan, Zhejiang)	Radio-frequency CNTFETs with a Y-shaped gate: f_T = 551 GHz and f_{max} = 1,024 GHz at 35 nm gate length

Two features of this table matter. First, the RF result: nanotube transistors are now among the fastest field-effect transistors of any material, because RF amplifiers need high transconductance and low capacitance but tolerate variability far better than logic does. Second, the TSMC result: a leading silicon foundry has built CNTFETs in the same gate-all-around format as its own 2 nm transistors. TSMC itself names the remaining problems: interface traps and channel variability. Its swing of 135 mV/dec shows how far the gate stack still is from the ideal.

Check your understanding

1. What does DREAM do, and why was it needed for RV16X-NANO?
2. Why are nanotubes suited to building transistors above finished silicon circuits?
3. Why have RF CNTFETs advanced faster than logic CNTFETs?

9 Where CNTFETs stand in 2026

> **Key ideas**
>
> As single devices, nanotube transistors match or beat silicon at the same dimensions.
>
> As a technology, they are where silicon was decades ago: thousands of transistors per circuit, not billions.
>
> The most likely first uses are in places silicon cannot reach easily: transistors built above silicon in 3D, radio-frequency front ends, and flexible or sensing electronics.

	Silicon GAA nanosheet (2 nm class, 2025–26)	Aligned-array CNTFET (best research, 2024–25)

	Silicon GAA nanosheet (2 nm class, 2025–26)	Aligned-array CNTFET (best research, 2024–25)
Channel	3–4 stacked Si sheets, 5 nm thick	100–200 tubes per µm, 1.2–1.8 nm diameter
On-current	≈ 1–2 mA/µm	1–3 mA/µm reported
Subthreshold swing	≈ 65–70 mV/dec	70–135 mV/dec in arrays (≈ 60–90 in single tubes)
n and p symmetry	Holes weaker; strain needed	Symmetric bands; contact-set polarity
Variability	Controlled at the 1 mV level	Tube count, type, diameter; traps
Largest circuit	≈ 10^{11} transistors per chip	≈ 1.5×10^4 transistors (2019)
Thermal budget	> 1000 °C front end	< 400 °C: can be built above silicon

The remaining problems, in roughly the order they must be solved:

1. **Uniform arrays.** Pitch and diameter variation must come down to the level where every transistor has the same number of equivalent tubes.
2. **Gate stacks.** Interface traps must fall so arrays switch at 60–70 mV/dec, as single tubes already do.
3. **Contacts at short length.** End-bonded or doped contacts must deliver low resistance on both n-type and p-type devices in a 10–15 nm contact.
4. **Doping and threshold control.** Stable, tunable thresholds for n-type and p-type devices across a wafer.
5. **Integration.** Thousands of transistors to billions: yield, design rules, models and tools.

The old notes ended in 2007 with "CNT is a future replacement for semiconductor-based microelectronics". Nearly twenty years later the honest statement is narrower: the physics case has been made in full, and the engineering case is being made one problem at a time.

Check your understanding

1. Which of the five remaining problems do you think is hardest, and why?
2. Give two applications where CNTFETs could be used before they can replace silicon logic.

10 Summary

- CNTFETs operate as **Schottky-barrier** transistors (the gate thins the contact barriers; ambipolar, worse swing, strongly diameter-dependent leakage) or as **MOSFET-like** transistors (doped contacts; the gate lowers a channel barrier; unipolar, near 60 mV/dec).
- A wrap-around gate gives $C_{ins} = 2\pi\varepsilon/\ln[(2t_{ox} + d)/d]$. In series with it is the **quantum capacitance** $C_Q = 8e^2/hv_F \approx 0.36$ fF/µm. Thin high-κ gates reach the quantum-capacitance limit.
- The **top-of-the-barrier model** gives the ballistic I–V: about 60 mV/dec, early saturation, about 40 µA per tube at 0.5 V; each subband carries at most $4q^2/h = 155$ µA/V.

- **Contacts** set the practical limit: 6.45 kΩ per tube is unavoidable; side contacts degrade below about 20 nm; end-bonded contacts do not.
- Polarity is set by **contact metals** (Pd for p, Sc for n) and electrostatic or dielectric doping; encapsulation is essential.
- Gate lengths of 5–10 nm work. The story since 2013 is integration: from 178 transistors to 14,702 (RV16X-NANO, 2019), commercial fabs (2020), wafer-scale arrays (2020), and 2024–25 results from TSMC and Peking University.
- In 2026 the obstacles are uniformity, gate-stack traps, short contacts, threshold control and scale.

Corrections to the old notes

Old note	What it said	Correction
Dandeliya & Srivastava (Gwalior)	Gate capacitance formula with r_{cnt} called the "nanotube diameter"	It is the radius; $C_{ins} = 2\pi\varepsilon / \ln[(t_{ox} + r)/r]$
Dandeliya & Srivastava	Comparison table: "Mobility 7×10^6 cm/s" (MOSFET) vs 3.5×10^6 cm/s (CNTFET), and "thermal velocity 2× MOSFET"	Those are velocities, not mobilities, and the two rows contradict each other
Dandeliya & Srivastava	"Von Neumann boundary conditions"	Neumann boundary conditions (zero normal field)
Dandeliya & Srivastava	"Increasing CNT diameter has good effect"	More on-current but much more off-current and ambipolar leakage; the useful window is about 1.2–1.8 nm
Dandeliya & Srivastava	Dielectric choice "not related to … passivating surface states"	Dielectric traps cause hysteresis, V_T spread and poor swing
Ragheb (Rice, 2007)	"Bachthold et al., Science 294, pp. 49–52"	Bachtold *et al.*, *Science* 294, 1317 (2001); pp. 49–52 is Tans *et al.* in *Nature*
Ragheb	Methods to "convert metallic CNTs into semiconductor type"	Plasma and electrical breakdown remove or damage metallic tubes; they do not convert them
Ragheb	"CNT is a future replacement for semiconductor based microelectronics"	Unproven as of 2026; see Section 9

Words to know

Term	Meaning
Schottky-barrier CNTFET	A CNTFET with metal contacts, switched by thinning the contact barriers
MOSFET-like CNTFET	A CNTFET with doped extensions, switched by a barrier in the channel
Ambipolar	Conducting both electrons and holes, depending on gate voltage
Quantum capacitance	$C_Q = e^2 D(E_F)$; the capacitance of filling the tube's own states
Quantum-capacitance limit	$C_{ins} \gg C_Q$: the bands follow the gate one-for-one

Term	Meaning
Top-of-the-barrier model	A ballistic model of a transistor from the charge and current at the barrier top
End-bonded contact	A metal (carbide) contact bonded to the end of the tube
Electrostatic doping	Making a region n^+ or p^+ with a gate instead of dopant atoms
Monolithic 3D	Building device layers on top of finished circuits on the same wafer
DREAM, RINSE, MIXED	The design and process methods behind the RV16X-NANO processor

In-class animations

Both at faizverse.github.io/nanoelectronic-devices/L4.

- **Ballistic CNTFET.** The top-of-the-barrier model of Section 4. Change the diameter, insulator thickness and κ, temperature and drain voltage, and watch the transfer and output curves and the quantum-capacitance ratio. Try thinning the insulator until the on-current stops rising.
- **Schottky-barrier CNTFET.** The band diagram along the tube with the tunnelling current of Section 4.4. Move the gate and see the barriers thin; change the contact metal and the diameter and watch the ambipolar curve change; switch to doped contacts and compare.

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